

Software Engineering and Project Management

LECTURE 2-3

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Part I

SOFTWARE PROCESSES

What some people think
Software Engineering is?

*Software Engineering is a great combination
between **artistry** and **engineering***

- Bill Gates

What is Software Engineering?

Software Engineering is about
processes to create software.

Concerned with theories, methods and
tools.

SE is Engineering

- **Making things that work** (*practical*)
- **Use of** models, standard designs, methods, etc.

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- **Constrains**
 - Time
 - Money
 - Organization

SE is Engineering

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- **Constrains**
 - Time
 - Money
 - Organization
- **Managing**
 - People
 - Communication

Questions about SE

Lets try to answer together ...

Attributes
of good
Software

SE vs.
Computer
Engineering
/System
Engineering

Fundamental
SE activities

Challenges
of Software
Engineering

Best SE
technique
s &
methods

...but not always like Engineering

- Every project is mostly new
- Software is “invisible”
- Perceived to be adaptable (*rewrite code vs. rebuild bridge*)
- Complexity
 - Lacks physical boundaries
 - Often does many things
 - Multi-language, multi-level
- Usability (*getting it right*)
- Legacy systems

Process

- **Process** is a series of actions (*in SE: activities*) that we take in order to achieve a result.
 - Planning is implicit in the definition
 - Activities is a placeholder for another process or a concrete action
- **Processes can be standardized**
 - Process model: abstraction, common themes
- **Problems with processes**
 - what really happens... too messy
- **A “methodological” process** cares of what *should* happen

Process-based approach to **Quality**

- **Assumptions**

- We **can** measure the **quality of our product**

- **Challenge**

- We **cannot** measure the **quality of a process**

- **Opportunity**

- cares of what **should** happen

**Key
Slide!**

Process-based approach to Quality



Product Quality

Cannot be measured
directly!

- **Acceptable**
 - usable, learnable, compatible
- **Dependable**
 - safe, reliable, secure
- **Efficient**
 - response time, memory use
- **Maintainable**
 - documented, structured

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**DIRECT
MEASURE**

Software Process Activities

- **Validation** *Testing*
- **Evolution** *Delivery,
Maintenance*
- **Specification** Requirements
- **Development** *Design, Implementation*

Q: In what order should these processes be followed?

Software Process Activities

- **Specification** *define functionality and constraints*
- **Development** *design and program*
- **Validation** *ensure it is what customer requires*
- **Evolution** *evolve to meet changing customer & market requirements*

**Key
Slide!**

Software Issues

Heterogeneity

Business &
Social
change

Security & trust

Software engineering diversity

- The SE methods and tools depend upon:
 - Type of application being developed
 - Requirements of the customer
 - Background of the development team

Applications Types

Stand-alone
applications

Data collection
systems

Interactive
transaction-based
applications

Systems of
systems

Embedded
control
systems

Batch
processing
systems

Entertainment
systems

Process descriptions

- Process descriptions also include:
 - Products --- outcomes of a process activity;
 - Roles --- responsibilities of the people involved in the process;
 - Pre- & post-conditions --- statements that are true before and after a process activity

Types of Software Processes

- **Plan Driven**

- *Process activities planned in advance*
- *Progress measured against plan*
- (Processes where all of the process activities are planned in advance and progress is measured against this plan.)

- **Agile**

- *Planning is incremental*
- *Early feedback from clients / users*
- (Processes where planning is incremental and it is easier to change the process to reflect changing customer requirements.)

- There are no right or wrong software processes.

Types of Software Processes

	Software Process Model	Advantages	Disadvantages
Plan Driven	• Waterfall • Incremental Development • Iterative development • Spiral Development • Prototype Model • Rapid Application Development	• Suitable for large systems and teams. • Handles highly critical systems effectively. • Appropriate for stable development environment. • Require experienced personnel at the beginning. • Success achieved through structure and order.	• Longer length in each iteration or increment. • Cannot accommodate changes any time. • Lack of user involvement throughout the life cycle of the product. • Costly for the dynamic development environment. • Assume that, future changes will not occur.
Agile	• Scrum model • Extreme Programming (XP) • Dynamic System Development Method • Kanban • Feature Driven Development	• Suitable for small to medium systems and teams. • Can accommodate changes at any time. • Effective for the dynamic development environment. • Required expert agile personnel throughout the life cycle. • Success achieved through freedom and chaos.	• Not suitable for large systems (except FDD). • Shorter length in each iteration. • Can accommodate changes at any time. • Costly for the stable development environment. • Assume that, frequent future changes will occur.

Questions



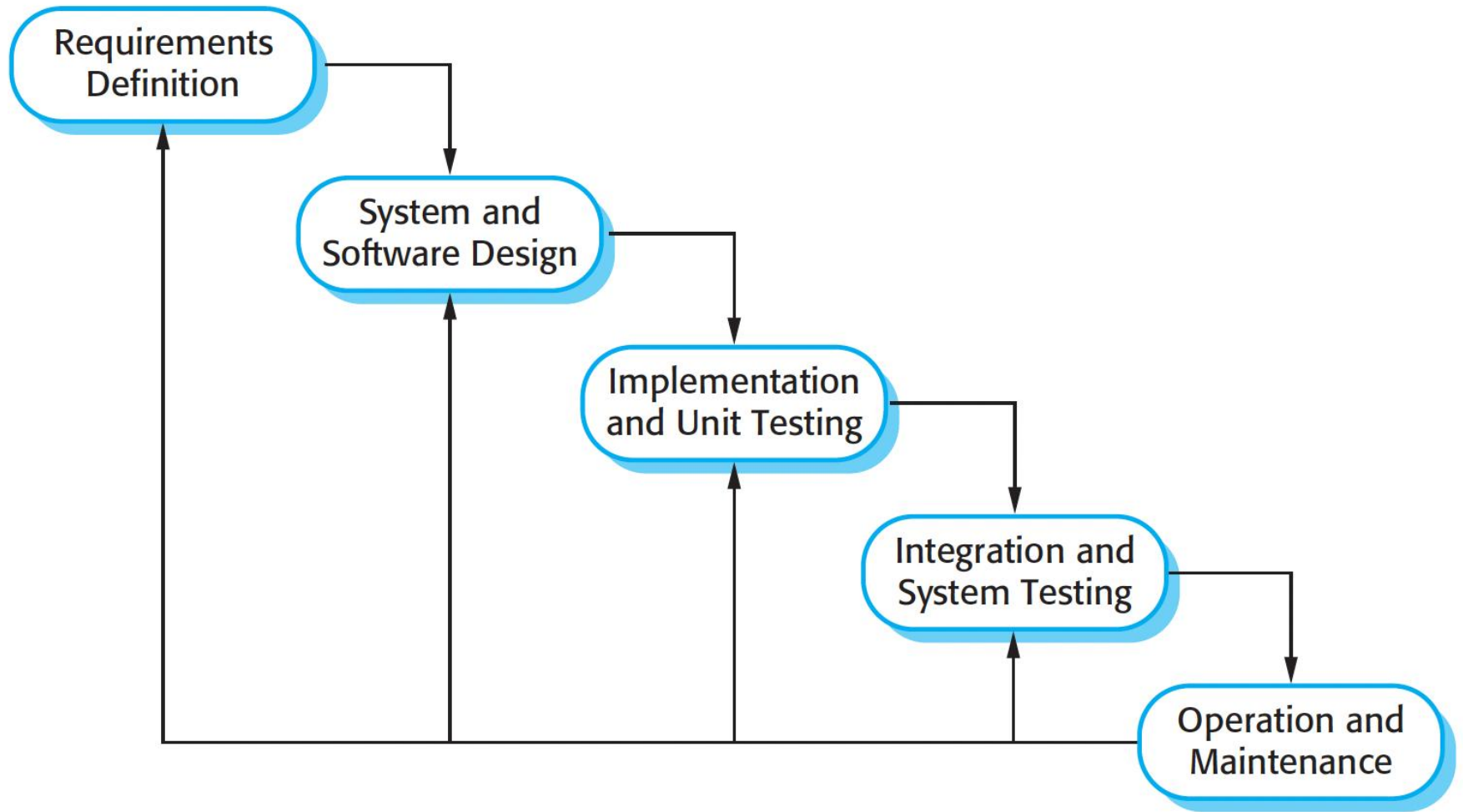
Part I.b

MODELS OF PLAN-BASED SOFTWARE PROCESSES

Software **process models**

- **The waterfall model**
- **Incremental development**
- **Reuse-oriented software engineering**

The **waterfall** model



Waterfall model

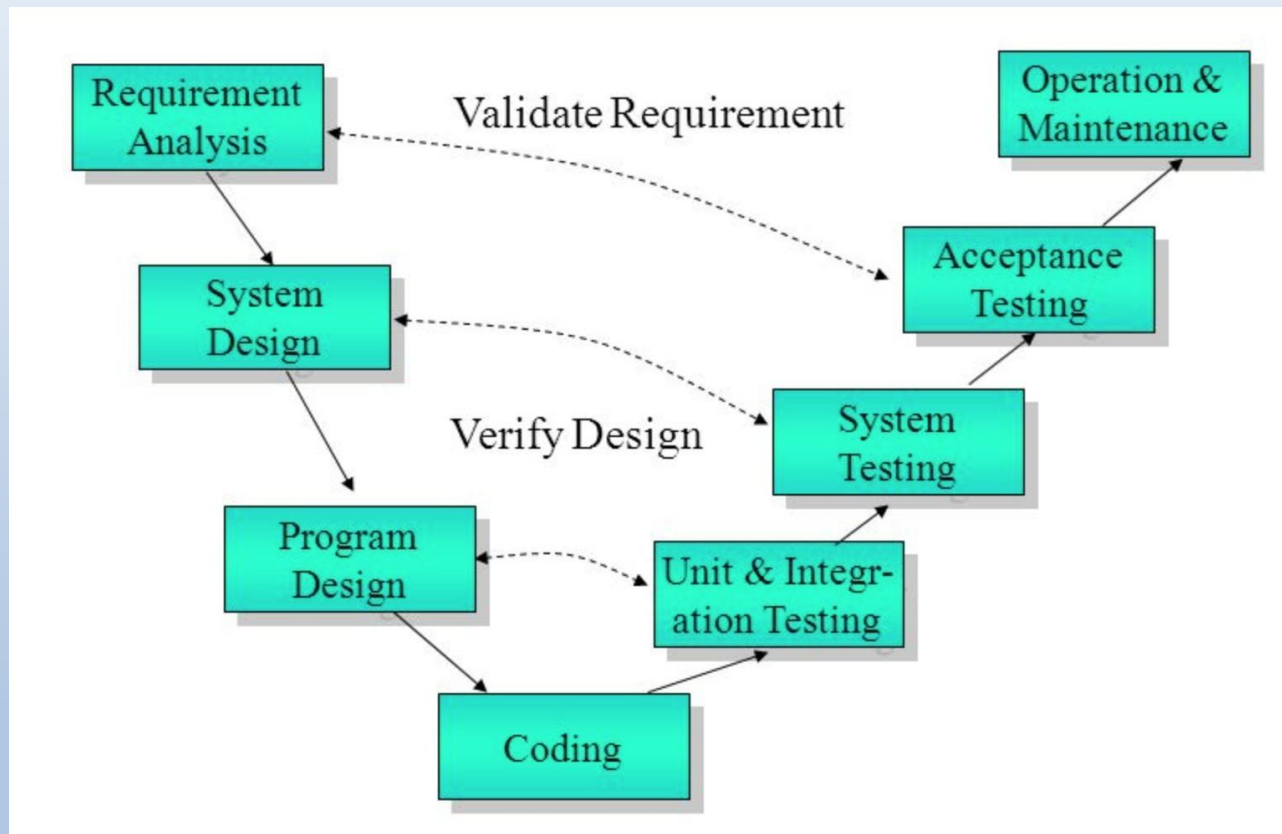
There are separate identified phases in the waterfall model:

- Requirements analysis and definition
- System and software design
- Implementation and unit testing
- Integration and system testing
- Operation and maintenance

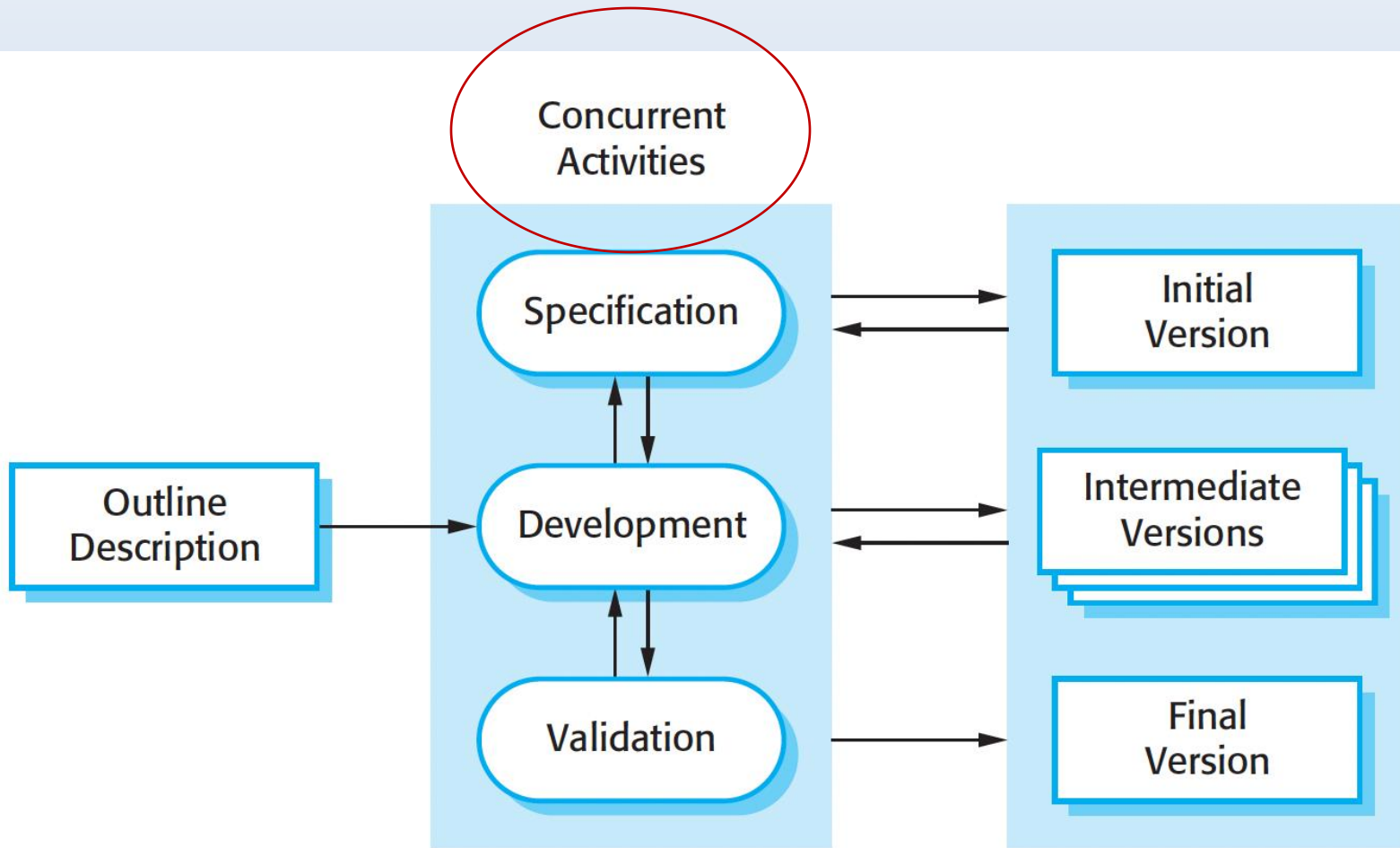
Waterfall model problems

- Inflexible: difficult to respond to changing customer requirements.
 - Appropriate only when the requirements are well-understood, and changes will be fairly limited during the design process.
 - Few business systems have stable requirements.
- The waterfall model is mostly used for large systems engineering projects where a system is developed at several sites.

The V-Model



Incremental development Model



Incremental Development

Pros & Cons

- The cost of accommodating changing customer requirements is reduced.
- It is easier to get customer feedback on the development work that has been done.
- .
- More rapid delivery and deployment of useful software to the customer is possible.

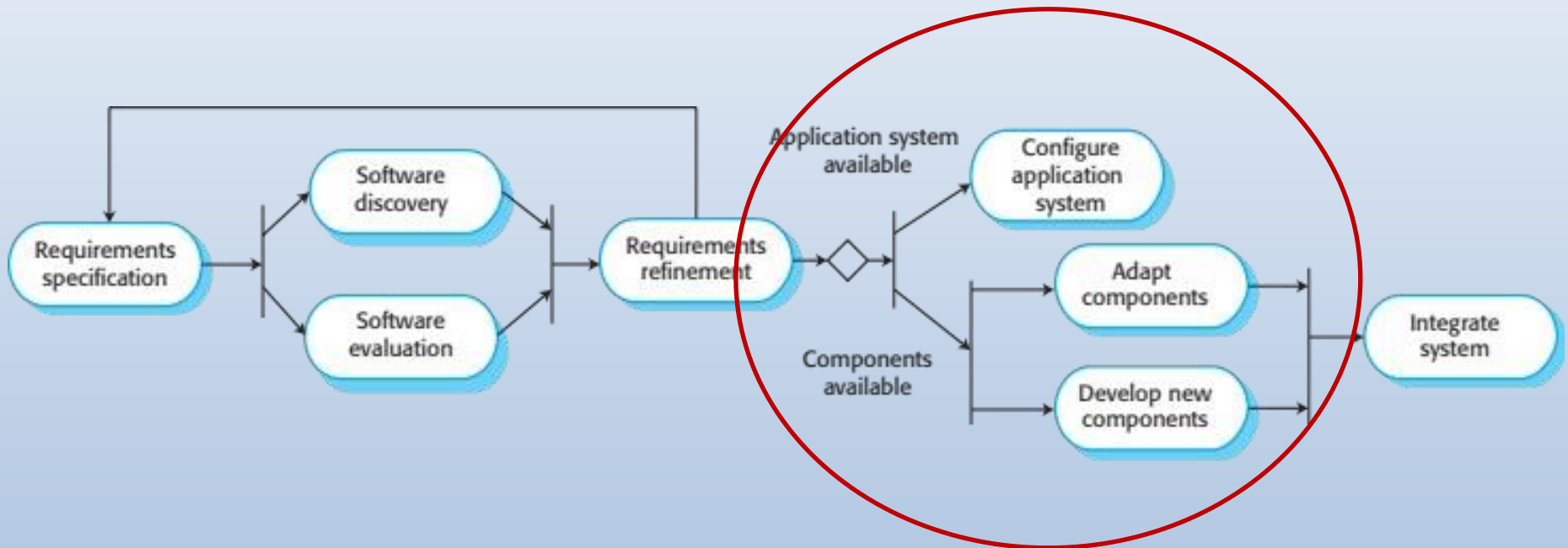
Incremental Development

Pros & **Cons**

- The process is not visible.
- System structure tends to degrade as new increments are added.

- Based on systematic reuse where systems are integrated from existing components or COTS (Commercial-off-the-shelf) systems.
- Process stages:
 - Component analysis
 - Requirements modification
 - System design with reuse
 - Development and integration.

Reuse-based Development



References

1. Ivar Jacobson and Ed Seidewitz. 2014. A new software engineering. Commun.ACM 57, 12 (November 2014), 49-54. DOI=10.1145/2677034
2. Cohen, M. An Iterative Waterfall Isn't Agile. Available at <https://www.mountaingoatsoftware.com/blog/an-iterative-waterfall-isnt-agile>

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Chapters	1.1, 1.2, 2.1, 2.2.3	1. Professional software development 2. Software engineering ethics, 2.1 Software process models, 2.2.3 Coping with change
Important Sections	2.1.1, 2.1.2, 2.2.3	1. Waterfall model, 2. Incremental development, 2.2.3 Software validation

Questions

